Homework 2: Traffic Light Controller FSM Design

Fall 2026 Embedded Systems (CEG-7360-0)

Q.1. Design a **Moore and Mealy FSM** that receives a **serial input `x`** and produces an output **`y = 1` whenever the sequence `101` is received/detected**. Otherwise, `y = 0`.

Show the following:

- The **block diagram** of the sequence detector.
- The **state diagram** for the **Moore FSM**.
- Convert the Moore FSM to a **Mealy FSM** and show the modified state diagram.
- Show the **hardware implementation using one-hot encoding**.
- Analyze the FSM operation using a **test input sequence** where `x = 101` is received, and show

Q.2 Use the **traffic-light FSM diagram in Slide 12** from the class lecture on **Finite State Machines: Moore and Mealy FSMs**.

1. Modify the state diagram by adding a **counter/timer** to keep the yellow-light state active for **T = 4 clock cycles**.
2. Draw the **block diagram** for the FSM, including the **inputs and outputs**.
3. Draw the **modified state diagram**, showing the **timer condition** and all inputs and outputs.
4. Show the **counter operation** for the yellow-light state for **4 clock cycles**.
5. Using the **one-hot controller concept discussed in class**, draw the **hardware implementation** of the traffic-light controller, showing the **FSM/controller, counter/timer, inputs, and outputs**.